OTEL-01: OpenTelemetry Fundamentals

> **Series:** OTEL — OpenTelemetry Integration | **Notebook:** 1 of 8 | **Created:** January 2026 | **Last Updated:** 10/13/2026

Introduction to OpenTelemetry and Dynatrace

OpenTelemetry (OTel) is the industry-standard framework for collecting telemetry data. Dynatrace fully supports OpenTelemetry through native OTLP ingestion, allowing you to leverage OTel instrumentation while benefiting from Dynatrace's AI-powered analytics — on its own, or combined with OneAgent per workload.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with OTLP ingestion enabled
Permissions	Platform token: `openpipeline:traces:ingest`, `openpipeline:metrics:ingest`, `openpipeline:logs:ingest`. Classic access token: `openTelemetryTrace.ingest`, `metrics.ingest`, `logs.ingest`
Knowledge	Basic observability concepts

1. What is OpenTelemetry?

OpenTelemetry is a vendor-neutral, open-source observability framework that provides:

Component	Purpose
APIs	Standardized interfaces for instrumentation
SDKs	Language-specific implementations
Collector	Agent for receiving, processing, and exporting telemetry
Protocol (OTLP)	Standard wire protocol for telemetry data

CNCF Project

OpenTelemetry is a Cloud Native Computing Foundation (CNCF) project, formed from the merger of OpenTracing and OpenCensus.

Key Benefits

Benefit	Description
Vendor Neutral	Open standard — the same instrumentation works with any OTLP-compatible backend
Standardized	Consistent across languages and platforms
Comprehensive	Traces, metrics, and logs in one framework
Community-Driven	Active development, broad adoption

2. OpenTelemetry and OneAgent in Dynatrace

OpenTelemetry and OneAgent are complementary, not alternatives. OpenTelemetry is the open, vendor-neutral foundation: one instrumentation standard and common semantic conventions across languages, environments and tools. OneAgent adds automation and runtime depth on top of that foundation. Dynatrace analyzes the two together *"regardless of how you mix your instrumentation choices"*, so the choice is made per workload rather than once for the whole organization.

What Each Adds

Aspect	OpenTelemetry	Dynatrace OneAgent
Role	Open telemetry foundation — APIs, SDKs, semantic conventions, OTLP	Automation and runtime depth on top of that foundation
Instrumentation	Per-runtime auto-instrumentation, plus SDK calls for custom spans	Automatic — injected at process start, no code change
Deployment	SDK and/or Collector, owned by your teams	Single agent with a centrally managed lifecycle
Coverage	The signals you choose to emit	Full stack — host, process, runtime, code level, topology
Portability	Any OTLP-compatible backend, Dynatrace included	Dynatrace, where its data is analyzed alongside OTel data
Dynatrace Intelligence	Yes — including service detection and service topology from OTel spans	Yes — with infrastructure topology filled in automatically
Runtime-access capabilities	Not provided by OTel telemetry	Application Security, Live Debugger, automatic RUM JavaScript injection

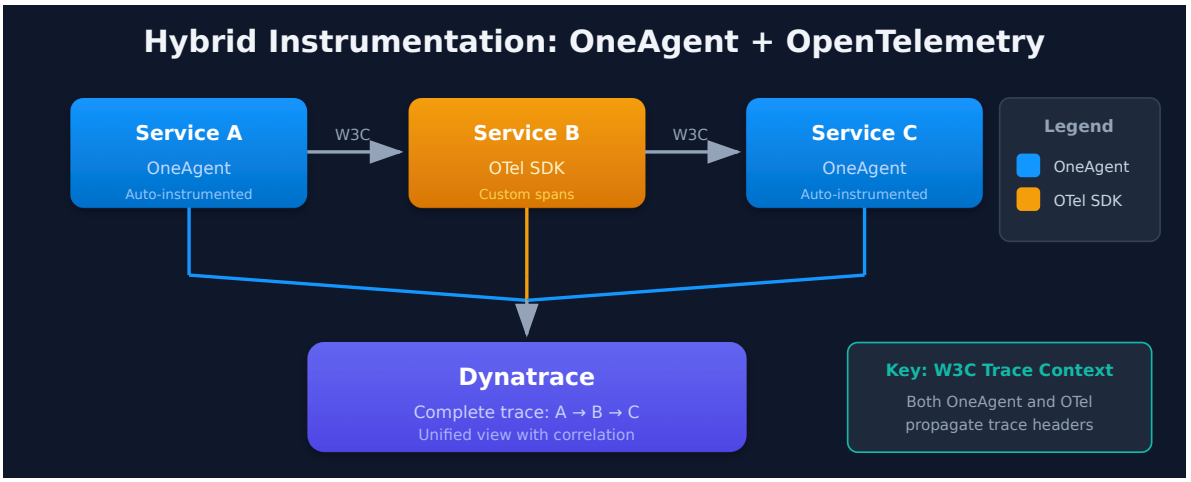
Choosing per Workload

Scenario	Recommendation
Organization has chosen OpenTelemetry	OTel — a complete path; add OneAgent only where its automation or runtime-access capabilities are wanted
No preference, standard runtimes on hosts, VMs or Kubernetes	OneAgent — fully managed, lowest effort; OTel for custom business spans
Existing OTel instrumentation	Keep it; add OneAgent alongside where runtime depth is wanted (see FAQ-03)
Prometheus endpoints	OTel Collector — the documented recommendation for new deployments

Scenario	Recommendation
Open-standard or multi-backend instrumentation mandate	OTel owns application instrumentation; OneAgent alongside where it runs
AWS Lambda	Dynatrace Lambda layer (Python, Node.js, Java, .NET); OTel for custom spans
Azure Functions	OneAgent on the plans and runtimes OneAgent 1.343+ supports; OTel elsewhere
GCP Cloud Functions, Ruby application code	OTel for application spans; OneAgent for host and process where it runs
Application Security, Live Debugger or RUM auto-injection needed	OneAgent on those workloads
Minimal effort	OneAgent

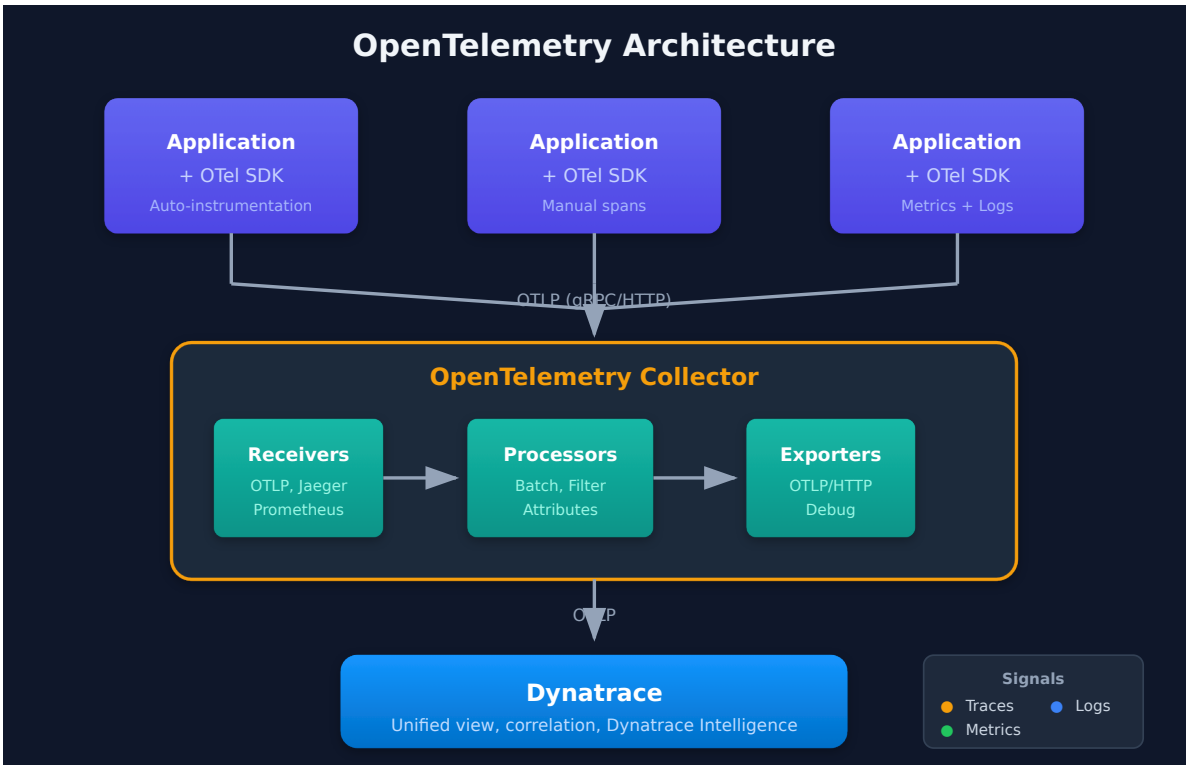
Using Both Together

Dynatrace supports using OneAgent and OTel together — in the same application, or across teams and workloads in the same environment:



3. Core Components

OpenTelemetry Architecture



Component Descriptions

Component	Role
API	Defines how to instrument (language-specific interfaces)
SDK	Implements the API, handles sampling, batching
Exporter	Sends data to backends (OTLP, Jaeger, etc.)
Collector	Standalone service for processing telemetry
Instrumentation Libraries	Pre-built instrumentation for frameworks

4. Signal Types: Traces, Metrics, Logs

Traces

Distributed traces track requests across services.

Concept	Description
Trace	End-to-end transaction path
Span	Single operation within a trace
SpanContext	Propagated context (trace ID, span ID)
Attributes	Key-value metadata on spans

Concept	Description
Events	Timestamped annotations on spans

Metrics

Quantitative measurements over time.

Instrument Type	Use Case	Example
Counter	Monotonically increasing value	Request count
UpDownCounter	Value that goes up and down	Active connections
Histogram	Distribution of values	Response time
Gauge	Current value	Temperature, queue size

Logs

Structured event records correlated with traces.

Field	Description
Timestamp	When the event occurred
Severity	Log level (DEBUG, INFO, ERROR)
Body	Log message content
Attributes	Structured metadata
TraceContext	Link to active span

5. Semantic Conventions

Semantic conventions define standard attribute names for consistent telemetry.

> **Note:** OpenTelemetry semantic conventions are evolving. HTTP conventions were **stabilized in 2023** with new names (see migration tables below). Database conventions are also migrating. SDKs support a gradual transition via the `OTEL_SEMCONV_STABILITY_OPT_IN` environment variable. Both old and new names may coexist during migration.

Resource Attributes

Attribute	Example	Description
`service.name`	`checkout-api`	Logical service name
`service.version`	`1.2.3`	Service version
`service.namespace`	`shop`	Groups services (team, system or application); scopes `service.name`
`deployment.environment.name`	`prod`	Deployment environment

> **Note:** `deployment.environment` was renamed to `deployment.environment.name` in the stable resource semantic conventions.

HTTP Span Attributes

Old (Legacy)	Stable	Description
`http.method`	`http.request.method`	HTTP method
`http.url`	`url.full`	Full URL
`http.status_code`	`http.response.status_code`	Response status code
`http.route`	`http.route` (unchanged)	Route template
`http.target`	`url.path` + `url.query`	Request target
`http.host`	`server.address`	Server hostname

> **Tip:** Use `OTEL_SEMCONV_STABILITY_OPT_IN=http` to emit only stable names, or `http/dup` to emit both old and new during migration. See the HTTP semconv migration guide.

Database Span Attributes

Old (Legacy)	Stable	Description
`db.name`	`db.namespace`	Database name/namespace
`db.statement`	`db.query.text`	Query statement
`db.operation`	`db.operation.name`	Operation type
`db.system`	`db.system`	Database type (unchanged)

> **Tip:** Use `OTEL_SEMCONV_STABILITY_OPT_IN=database` for stable database names. See the DB semconv migration guide.

Kubernetes Attributes

Attribute	Example	Description
`k8s.namespace.name`	`checkout`	Namespace
`k8s.pod.name`	`checkout-api-abc123`	Pod name
`k8s.deployment.name`	`checkout-api`	Deployment

6. Dynatrace OTel Integration

OTLP Endpoints

Dynatrace accepts OTLP data natively via **HTTP only**, with binary Protocol Buffers: *"gRPC is not supported. API calls need to use HTTP."* and *"JSON is not supported for Protocol Buffers. Binary format must be used."*

Protocol	Endpoint	Notes
HTTP	`https://{your-env}.live.dynatrace.com/api/v2/otlp`	Recommended for direct ingest
gRPC	Not supported for direct Dynatrace ingest	Use a Collector to convert gRPC → HTTP

> **Important:** If your application uses gRPC to export OTLP data, route it through an OpenTelemetry Collector configured with an `otlp` gRPC receiver and an `otlphttp` exporter to Dynatrace. See Transform OTLP gRPC.

Authentication

Two token types work, with different schemes and scopes:

Token	Header	Traces	Metrics	Logs
Platform token	`Authorization: Bearer <token>`	`openpipeline:traces:ingest`	`openpipeline:metrics:ingest`	`openpipeline:logs:ingest`
Classic access token	`Authorization: Api-Token <token>`	`openTelemetryTrace.ingest`	`metrics.ingest`	`logs.ingest`

> **Sources:** OTLP API (DT docs) — *"Platform token: Use Bearer in the Authorization header. The required scopes are openpipeline:logs:ingest, openpipeline:metrics:ingest, and openpipeline:traces:ingest."*

Configuration Example

OTel Collector Exporter:

```
exporters:
  otlphttp:
    endpoint: https://{your-env}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
```

SDK Direct Export (Python):

```
import os

from opentelemetry.exporter.otlp.proto.http.trace_exporter import OTLPSpanExporter

exporter = OTLPSpanExporter(
    endpoint="https://{your-env}.live.dynatrace.com/api/v2/otlp/v1/traces",
    headers={"Authorization": f"Api-Token {os.environ['DT_API_TOKEN']}"}
)
```

OpenTelemetry Licensing & Cost

OpenTelemetry data is billed exactly like any other ingested data under the Dynatrace Platform Subscription (DPS) — there is no separate OTel licence and no OTel surcharge or discount. Each signal maps to DPS capabilities:

Signal	Ingest & Process	Retain	Query
Traces	per GiB ingested	GiB-days (10 days included)	per GiB scanned (Distributed Tracing / Services apps don't bill queries)
Metrics	per 100k data points	GiB-days (15 months @ 1-min granularity included)	included with Ingest & Process
Logs	per GiB ingested	GiB-days	per GiB scanned

Included volume follows the source, not the protocol: Full-Stack Monitoring includes a defined trace volume *"for traces that are sent via the OneAgent Trace API or from a Full-Stack monitored host"*, and *"Metrics originating from Full-Stack-monitored*

hosts or containers include a defined amount of metric data points" — you are charged only for data above the included amount. OTel telemetry from sources outside Full-Stack Monitoring (for example, a workload on a host with no OneAgent) has no included volume to draw on and bills on the rate card as ingested. Practical consequence: high-volume OTel traces/metrics from such sources are a direct cost line — size and (if needed) sample them deliberately (see OTEL-04 §7 and the FINOPS series).

> **Sources:** OpenTelemetry licensing (DT docs) — *"With Dynatrace, OpenTelemetry data is treated like any other ingested data."* **Derived:** OTel data from sources outside Full-Stack Monitoring has no included volume, because the page scopes the allowance to trace and metric data from those sources.

```
// View OpenTelemetry traces that arrived over OTLP
// dt.openpipeline.source identifies the ingest path. otel.scope.name is NOT a reliable
// "is OTel" test: OneAgent spans can carry an instrumentation scope too.
fetch spans, from:-1h
| filter dt.openpipeline.source == "/api/v2/otlp/v1/traces"
| fields start_time, trace.id, service.name, span.name, otel.scope.name, duration
| sort start_time desc
| limit 20
```

```
// OTel instrumentation libraries in use (OTLP-ingested spans)
fetch spans, from:-1h
| filter dt.openpipeline.source == "/api/v2/otlp/v1/traces"
| summarize count = count(), by:{otel.scope.name, otel.scope.version}
| sort count desc
| limit 20
```

7. Choosing the Combination per Workload

Where OpenTelemetry Leads

Scenario	Why OTel
OTel as the collection strategy	A complete path — services, endpoints and service topology are detected from OTel data
Custom business telemetry	Spans and metrics specific to your domain
Prometheus endpoints	Collector-based scraping is the documented recommendation for new deployments
Open-standard mandate	One instrumentation model across teams and backends
Multi-cloud / polyglot	Consistent instrumentation across clouds and languages
Where OneAgent does not run	GCP Cloud Functions, unsupported Azure Functions plans, Ruby application code
Edge/IoT	Lightweight collection

Where OneAgent Adds Depth

Scenario	Why OneAgent
Full-stack	Infrastructure, process, runtime and code in one install
Zero-effort	Automatic instrumentation, centrally managed updates
Infrastructure topology	Hosts, processes and containers in Smartscape without running a Collector
PurePath	Code-level transaction tracing
Application Security	Runtime Vulnerability Analytics and Runtime Application Protection
Live Debugger	Java and Node.js, OneAgent 1.309+

Scenario	Why OneAgent
Real User Monitoring	Automatic RUM JavaScript injection
Serverless (supported platforms)	AWS Lambda layer; supported Azure Functions plans from OneAgent 1.343

Recommended: Both, per Workload

For most enterprises:
- **OTel-only is a complete path** where OTel is the collection strategy — the tooling decision should not push teams off it
- **OneAgent** where teams want fully managed collection and the capabilities that need runtime access
- **Both** where a workload benefits from OTel's open instrumentation and OneAgent's depth together
- **Mixed estates are normal** — decide per team, application, host or Kubernetes workload

Next Steps

Now that you understand OTel fundamentals, proceed to:

Next Notebook	Topic
OTEL-02: Collector Architecture	Deep-dive into the Collector
OTEL-03: Collector Deployment	Deployment patterns
OTEL-04: Trace Instrumentation	Instrumenting your code

Summary

In this notebook, you learned:

- What OpenTelemetry is and its benefits
- How OpenTelemetry and OneAgent complement each other
- Core OTel components: API, SDK, Collector
- Signal types: traces, metrics, logs
- Semantic conventions for consistent telemetry (including the stable HTTP and DB naming migration)
- Dynatrace OTLP integration configuration (HTTP only — gRPC requires Collector)
- How to choose the OTel and OneAgent combination per workload

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References

- OpenTelemetry Official Docs
- Dynatrace OpenTelemetry Integration — *"regardless of how you mix your instrumentation choices"*
- Dynatrace OpenTelemetry Collector (Dynatrace blog, 03/18/2024) — *"Used together, Dynatrace extends OpenTelemetry observability, and OpenTelemetry extends Dynatrace observability."*
- Unified services (DT docs) — services detected with Service Detection v2 *"were first introduced for OpenTelemetry"*
- OpenTelemetry Host Monitoring extension (DT docs) — *"This extension automatically generates entities for hosts and their running processes."*
- Prometheus data source (DT docs) — *"For high-volume Prometheus scraping in Kubernetes, and for new deployments, consider the OpenTelemetry Collector"*

- Runtime Vulnerability Analytics (DT docs) — *"Dynatrace OneAgent monitors loaded libraries, runtime components, and data flows in real time."*
- Runtime Application Protection (DT docs) — *"Code-level visibility provided by OneAgent"*
- Live Debugger (DT docs) — *"OneAgent version 1.309+ with Java Live-Debugger and Node.js Live-Debugger"*
- RUM JavaScript injection (DT docs) — *"Once OneAgent is deployed in full-stack monitoring mode on a host, it by default automatically injects the RUM JavaScript"*
- AWS Lambda integration (DT docs) — *"using auto-instrumentation without code changes"*
- OneAgent 1.343 release notes (DT docs) — Azure Functions: Python on the Flex Consumption Plan for Linux; Java and Node.js on Consumption, Premium, and Dedicated plans for Windows
- OTLP Specification
- Semantic Conventions
- HTTP Semconv Migration Guide
- DB Semconv Migration Guide

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